RAMAIAH Institute of Technology

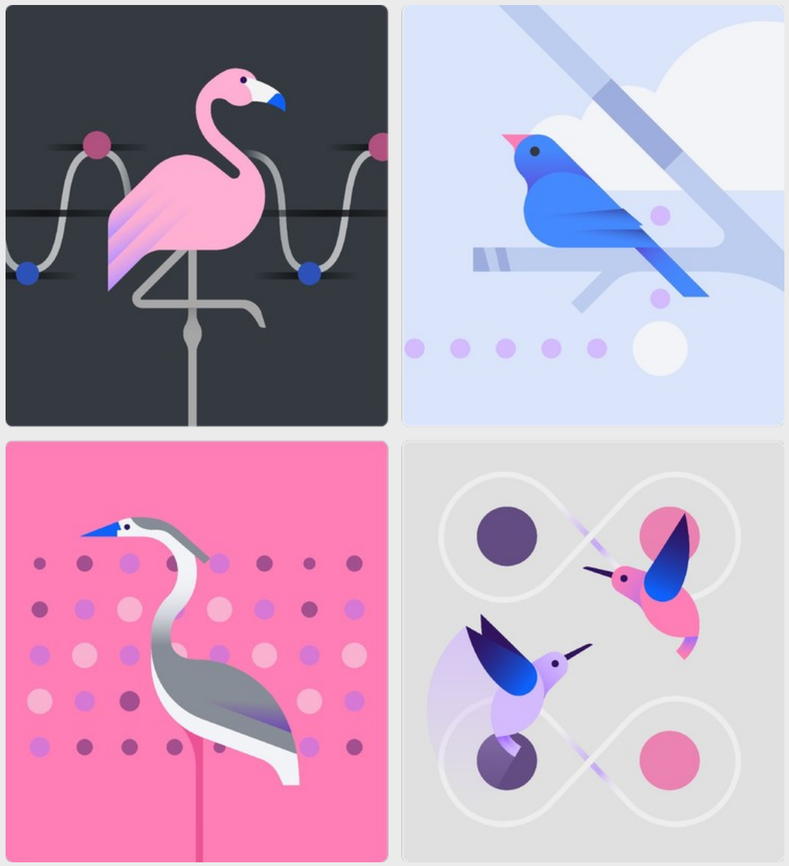
IEEE Computational Intelligence Society Bangalore Chapter

RIT

QUANTUM WEEK 26

IBM Quantum

Q-HACK INDIA 2026



Team Name:

Team Lead Name:

Track:

Problem Statement title:

Q-SHIELD: End-to-End Quantum Security Platform

Discover, understand, protect, and migrate quantum-vulnerable cryptography before Q-day

Problem

Today's public-key cryptography has an expiry date

1

Shor breaks RSA & ECC

A large fault-tolerant quantum computer running Shor's algorithm derives private keys from public keys in polynomial time. RSA, ECDSA, ECDH, and Diffie-Hellman all fall.

2

Harvest now, decrypt later

Attackers record encrypted network traffic and exfiltrate ciphertext today to decrypt once quantum hardware scales. Long-lived financial, healthcare, and state records are exposed already.

3

No inventory, no plan

Most organizations lack visibility into where vulnerable algorithms live across polyglot repositories, configurations, and certificates, making multi-year migrations difficult to sequence.

The clock is set: NIST standardized ML-KEM, ML-DSA, and SLH-DSA in August 2024 (FIPS 203–205) and plans to deprecate quantum-vulnerable public-key algorithms after 2030 (NIST IR 8547).

QUANTUM WEEK 26

IBM Quantum

Why It Matters

The deadline is set by your data, not by quantum computers

Who faces it

1. **Banks & payments**
   transaction records, card data, inter-bank clearing keys
2. **Healthcare**
   medical records and genomic data kept for decades
3. **Government & defense**
   classified intelligence, citizen identity registries
4. **Any organization**
   proprietary data that must stay secret for 10+ years

Mosca's inequality: data is at risk when

$$X + Y > Z$$

X = years data must stay secret · Y = years to migrate · Z = years until Q-day

Example from our scan: QuantumBank payment-api

records kept: 10 y	migrate: 2.3 y	Q-day ≈ 12 y

Result: 10.0 + 2.3 = 12.3 y > 12.0 y (Q-day)
RSA-protected financial records outlive their encryption today, rated high exposure. Migration has to start now.

Proposed Solution

One platform from discovery to migration

1 DISCOVER

Crypto scanner

Scans Python, Java, JS and Go code, SSH/JWT configs, X.509 certificates and committed keys into a Cryptographic Bill of Materials.

2 UNDERSTAND

Risk + quantum threat

Readiness score 0–100, harvest-now-decrypt-later exposure per data class, Shor and Grover simulated.

3 PROTECT

PQC gateway

Quantum-safe channel with ML-KEM-768 key exchange and ML-DSA-65 signatures, hybrid with X25519.

4 MIGRATE

Crypto-agility engine

A replacement for every finding, a 4-phase roadmap and the projected readiness after each phase.

5 AI SECURITY COPILOT: Explains the risk, answers "what should we migrate first?", writes developer tickets and an executive audit report, strictly grounded in verified CBOM finding IDs.

Why Quantum?

Quantum computing is both the threat and our method

The threat

Shor's algorithm [Broken]
Exponential speedup for factoring and discrete logarithms: RSA and elliptic-curve keys can be derived from public keys.

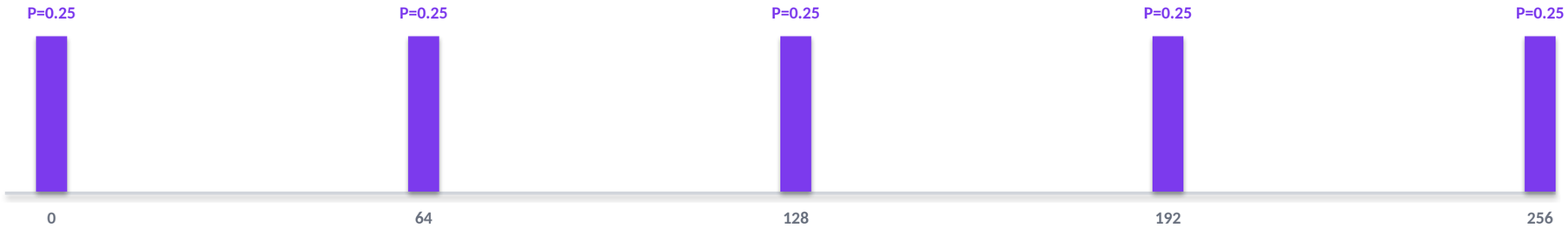
Grover's algorithm [Weakened]
Quadratic speedup for brute-force search: AES-128 drops to about 64-bit strength; AES-256 keeps 128 bits.

Our method

Exact statevector simulation of Shor's period finding: modular exponentiation f(x) = a^x mod N, inverse QFT, measurement, then continued fractions and gcd.

Post-quantum ML-KEM and ML-DSA are built on lattice problems that resist this adversary because they lack periodic abelian structure.

Shor simulation, N = 15, a = 7: peaks occur every Q/r = 64 (s in {0, 64, 128, 192})



Implementation and Architecture

One inventory feeds every module

Inputs

- Git repository URL
- ZIP archive upload
- Polyglot source code
- Server configurations
- Certificates & keys

FastAPI backend (Python)

Crypto scanner
Polyglot static analysis with comment stripping

Risk engine
Mosca inequality & Grover degradation calculation

Quantum simulator
Exact statevector Shor & multi-qubit circuits

PQC gateway
Hybrid ML-KEM-768 + ML-DSA-65 channel

Migration engine
Contextual code replacement diff generator

AI copilot
Grounded advisor validated against SQLite CBOM

Cryptographic Bill of Materials (CycloneDX 1.6 Schema) · SQLite

Outputs

- Guided console (5 stages / 7 tabs)
- Quantum-safe encrypted channel
- Executive risk audit report (Markdown)
- Developer tickets (Jira / GitHub)
- Machine-readable CBOM export (JSON)

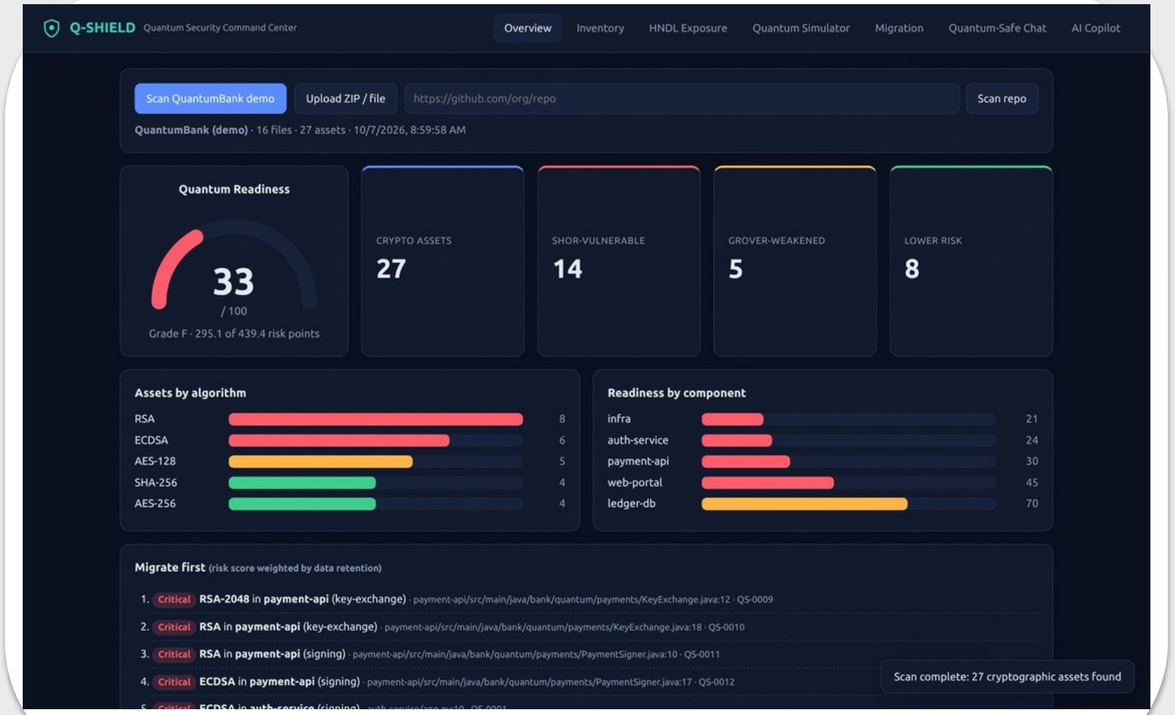
Stack: Python · FastAPI · pyca/cryptography (OpenSSL 3.5) for ML-KEM-768, ML-DSA-65, AES-256-GCM · NumPy statevector · SQLite · Plain JS + SVG · Docker

QUANTUM WEEK 26

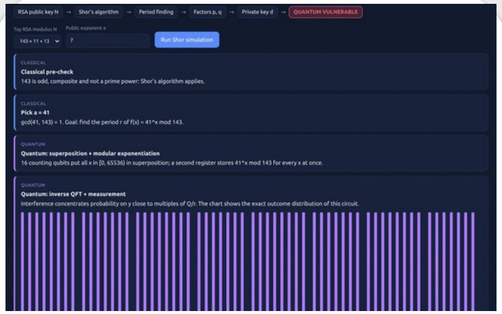
IBM Quantum

Minimal Prototype

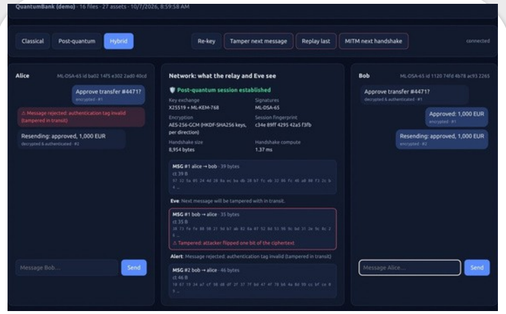
A working platform, not a mock-up



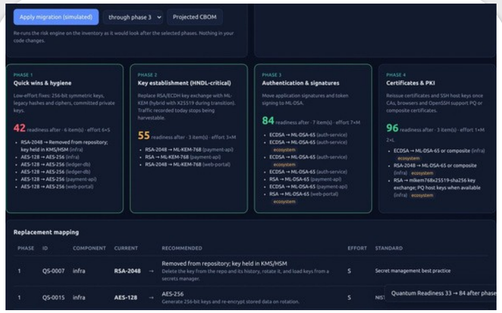
Overview: scan of QuantumBank, 27 assets cataloged, readiness evaluated at 33/100



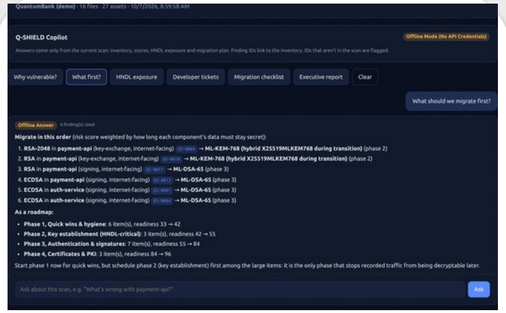
Shor simulation on 143



ML-KEM chat, tamper rejected



Roadmap: 33 → 84 → 96



Grounded AI copilot

Results / Validation

Validated on a seeded bank, end to end

27 / 27

seeded crypto assets found in QuantumBank, 0 hits on commented-out code or docs

33 → 84

readiness after roadmap phases 1–3 (96 after phase 4), recomputed by the risk engine

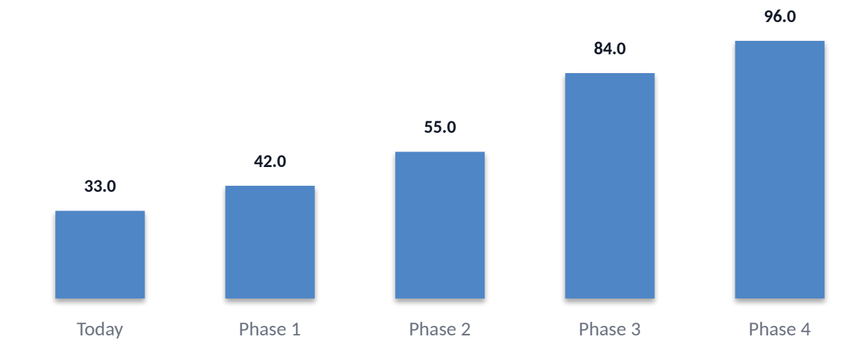
8,954 B

hybrid ML-KEM-768 + ML-DSA-65 handshake, a few ms of compute

84

automated tests passing, including tamper, replay and MITM rejection

Quantum Readiness by roadmap phase (QuantumBank)



Also validated

- Shor simulation factors 15, 21, 143, 221 and 247; peaks match theory exactly
- ML-KEM/ML-DSA wire sizes match FIPS 203/204 (1184 B key, 1088 B ciphertext, 3309 B signature)
- Copilot cites only real finding IDs; invented IDs are flagged and rejected
- ZIP-slip archives and unsafe Git URLs rejected

Roadmap

From demo app to real codebases

1

Deeper discovery

- AST and data-flow analysis
- Binaries and container images
- Live TLS/SSH endpoint scanning

2

Standards & CI

- CycloneDX CBOM export
- CI gate blocking vulnerable crypto
- Automated GitHub app remediation

3

Production PQC

- Hybrid TLS gateway (X25519MLKEM768)
- Post-quantum / composite certificates
- Cloud KMS and HSM integration

4

Real validation

- Precision/recall on public repos
- Enterprise pilot with a partner bank
- Shor N = 15 on IBM Quantum hardware

Round 2 goal: scan real repositories with measured accuracy, and run the period-finding circuit on real quantum hardware.

References and Sources

Standards, Research and Foundations

Standards & research

- NIST FIPS 203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA), Aug 2024.
- NIST IR 8547 (initial public draft), Transition to Post-Quantum Cryptography, Nov 2024
- P. W. Shor, Polynomial-time algorithms for prime factorization on quantum computer, 1997
- L. K. Grover, A fast quantum mechanical algorithm for database search, STOC 1996
- M. Mosca, Cybersecurity in an era with quantum computers: will we be ready?, IEEE 2018
- C. Gidney, M. Ekerå, Factor 2048-bit RSA in 8h using 20M noisy qubits, arXiv:1905.09749
- C. Gidney, Factor 2048-bit RSA with less than a million noisy qubits, arXiv:2505.15917

Software & documentation

- pyca/cryptography: ML-KEM, ML-DSA, AES-GCM, HKDF bindings
- FastAPI, NumPy, Pydantic, Uvicorn, SQLite
- Anthropic Claude & Google Gemini APIs (Security Copilot)
- CycloneDX 1.6 Cryptographic Bill of Materials (CBOM) specification
- Open Quantum Safe (OQS) project (architectural reference)

Originality Statement

Q-SHIELD is our team's original implementation engineered for Q-Hack India 2026. Foundational libraries are consumed as dependencies, not forked or copied.